Mid-tern, Catch-up

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https://github.com/hyunneee/hyunneee.github.io

in [VMP_midterm catch-up] branch

4. **A solar system with the sun, venus, earth, moon, and a rocket wondering around.**

```python
import numpy as np
import cv2

height , width = 1000, 1500

def getline(x0,y0,x1,y1):
    points = []
    if abs(x1-x0)>=abs(y1-y0):
        n = abs(x1-x0)
    else:
        n = abs(y1-y0)
    dx = (x1-x0)/n
    dy = (y1-y0)/n
    x,y = x0,y0
    for k in range(n):
        points.append((int(x),int(y),1))
        x = x+dx
        y = y+dy
    return points

def drawLine(canvas, x0, y0, x1, y1, color=(255, 255, 255)):
    xys = getline(x0, y0, x1, y1)
    for xy in xys:
        x, y, a = xy
        canvas[y, x, :] = color
    return

def drawPolygon(canvas, pts, color, axis=False):
    n = pts.shape[0]
    for k in range(n):
        drawLine(canvas, pts[int(k%n),0], pts[int(k%n),1],
                        pts[int((k+1)%n),0], pts[int((k+1)%n),1], color)

    if axis == True: # center - pts[0]
        center = np.array([0., 0, 0])
        for p in pts:
            center += p
        center = center / pts.shape[0]
        center = center.astype('int')
        # print(center)
        drawLine(canvas, center[0],center[1], pts[0][0],pts[0][1], color=(255, 128, 128))
    return
```

```python
def main():
    canvas = np.zeros( (height, width, 3), dtype='uint8')

    theta = 0
    theta2= 0
    theta3 = 0
    a,b = -750, -500

    while True:
        # get ngon points
        sun = getRegularNGon(40) # vertices of the N-gon
        venus = getRegularNGon(40)
        earth = getRegularNGon(40)
        moon = getRegularNGon(40)
        rocket = getRegularNGon(4)

        # scale up
        sun[:,0] *= 100
        sun[:,1] *= 100
        venus[:,0] *= 30
        venus[:,1] *= 30
        earth[:,0] *= 30
        earth[:,1] *= 30
        moon[:,0] *= 20
        moon[:,1] *= 20
        rocket[:,0] *= 10
        rocket[:,1] *= 10

        # rotating, translating points
        P = makeTmat(a,b) @ makeRmat(theta)
        S = P @ makeRmat(theta) @ makeTmat(-300,0)
        # M = S @ makeRmat(-theta) @ makeRmat(theta2) @ makeTmat(-50,0) @ makeRmat(-theta2)

        sun = P @ sun.T
        sun = sun.T
        sun = sun.astype('int')
        drawPolygon(canvas, sun,(50,0,255),axis = True)
        drawColored(canvas,sun,(50,50,255))
```

+theta 값 리셋

도형이동 + 회전 -

붉은색.

```python
        V = P @ makeRmat(theta) @ makeTmat(-180,0)
        venus = V @ venus.T
        venus = venus.T
        venus = venus.astype('int')
        drawPolygon(canvas, venus, (0,70,150),axis = True)
        drawColored(canvas,venus,(0,70,150))

        rocket = V @ makeRmat(-theta) @ makeRmat(theta2) @ makeTmat(-50,0) @ makeRmat(-theta2) @ rocket.T
        rocket = rocket.T
        rocket = rocket.astype('int')
        drawPolygon(canvas,rocket,(200,100,50),axis= True)
        drawColored(canvas,rocket,(255,255,255))

        E = P @ makeRmat(theta) @ makeTmat(-340,0)
        earth = E @ earth.T
        earth = earth.T
        earth = earth.astype('int')
        drawPolygon(canvas, earth, (163,103,0),axis = True)
        drawColored(canvas,earth,(163,103,0))

        moon = E @ makeRmat(-theta) @ makeRmat(theta3) @ makeTmat(-70,0) @ moon.T
        moon = moon.T
        moon = moon.astype('int')
        drawPolygon(canvas,moon,(128,128,128),axis= True) #purple
        drawColored(canvas,moon,(128,128,128))

        cv2.imshow("window", canvas)
        canvas[:, :, :] = 0
        theta+=1
        theta2 +=15
        theta3 += 8
        if cv2.waitKey(20) == 27: break

#
if __name__ == "__main__": # __
    main()
```

태양에서 거리 -180 주기만큼 떨어짐

갈색.

로켓 비너스 주위 공전. (theta2 속도만큼)

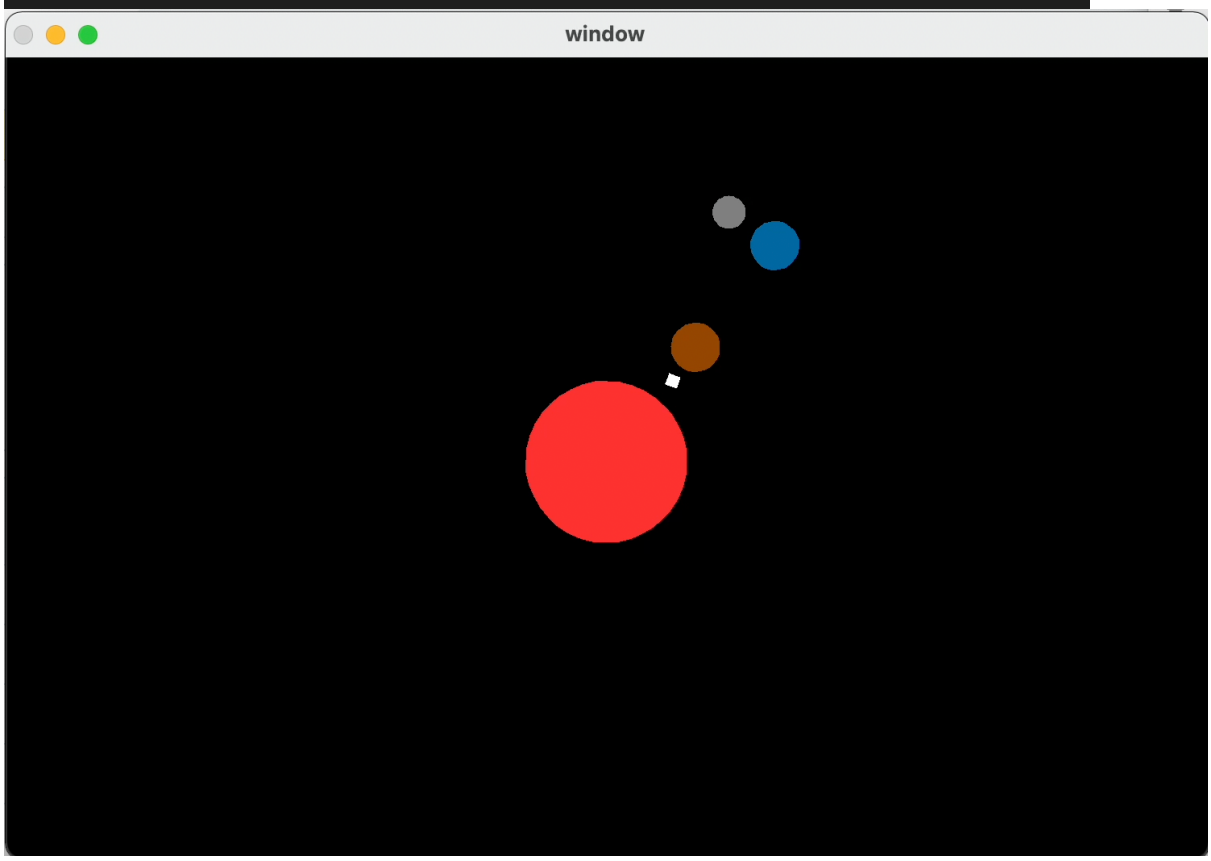
→ 흰색,

지구는 태양에서 거리 -340 ''

→ 파란색

달은 지구 주위 공전 (theta3 속도만큼)

→ 회색.

→ 지우기

theta 값으로 속도조절